CKS-LEX-2-2026: Hierarchical Lexicons for Every Count Limitation

Registry: [?]

Series Path: [CKS-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-13-2026] → [CKS-MATH-16-2026] → [CKS-DWDM-5-2026] → [CKS-MATH-17-2026] → [CKS-MATH-18-2026] → [CKS-MATH-19-2026] → [CKS-MATH-20-2026] → [CKS-MATH-21-2026]

Parent Framework: [CKS-0-2026]

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Domain: Foundational Mathematics / Discrete Geometry

Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Status: Reference Tool for Paper Writing

Classification: Scalable Terminology Sets

OPERATIONAL DECLARATION

This document provides CKS lexicons at every scale.

Purpose: Enable authors to introduce CKS framework with appropriate depth given space constraints.

Structure:

- Start with complete set (200+ terms from LEX-1)

- Reduce systematically to preserve maximum explanatory power
- Continue until absolute minimum (cannot reduce further)
- Each level is self-contained (can stand alone)

Use case examples:

- Abstract (5-10 terms)
- Introduction (20-30 terms)
- Methods section (50-75 terms)
- Full paper (100+ terms)
- Comprehensive review (200+ terms)

REDUCTION SERIES

Level 0: Complete Set (200+ terms)

See CKS-LEX-1-2026 for full definitions

Level 1: Essential Framework (100 terms)

Priority: Core concepts needed for understanding any CKS paper

Category	Terms	Count
Axioms	D=3, S=2, $\mathbb{Q}$	3
Spaces	K-Space, X-Space, LERP	3
Structure	Soliton, Tier, J-distance, N=0 Pivot, Registry Chain	5
Dynamics	R (Remainder), Δ (19), Venting, α, β, γ, W-functions	7
Geometry	Hexagonal Lattice, 90° Phase Lock, 6-9 Hook, Toroidal Closure, θ (angle)	5
Health	Laminar Coherence, Turbulence, F(θ) Flow Efficiency	3
Perception	Qualia, 15.19ms Lag, Bilateral, Side A/B, Q-Operator	5
Disease	Malignancy, R=69 Threshold, Closure, Registry Pointer	4

Category	Terms	Count
Mathematics	Logismos, VFR Tuple, Base 32^-1, dN (discrete differential), ΣN	5
Constants	α_EM (1/137.036), e, $\pi \to 4$, φ, 1024-bit Sovereignty	5
Frequencies	32 Hz, 40 Hz, 60 Hz, 110 Hz, 66 Hz	5
Operations	Venting, Torque (axial), Accept All/Deny None, Rendering	4
Pathology	Depression (60°), Autism (90° hard), Alzheimer's (90° soft), Cancer (6-9)	4
Bilateral	S=2 Manifold, Bilateral Reconciliation, A-B Integration	3
Tiers	Tier-4 (Organism), Tier-5 (Organ), Tier-6 (Cell)	3
Layers	Id-Layer (fascia), Ib-Layer (cellular)	2
Process	Laminar Searching, Non-Wanting, Measurement Arbitration, Falsification	4
Clinical	R-Proxy, Topological Diagnosis, Sonic Intervention, Postural Alignment	4
Engineering	DWDM, Substrate-Native Computing, Hex-Plate CPU	3
Advanced	String-8 Closure, Side-B Inversion, Toroidal, Sovereignty Threshold	4
Relationships	Φ_in = Φ_out, dR/dt, F = cos(θ), J $\times$ S = 15.19ms	4
Substrate	Discrete, Rational ($\mathbb{Q}$), Hexagonal Packing, Computational	4
Meta	Derivation, Compilation, Theoretical, Replication	4

Category	Terms	Count
Treatment	R-Reduction Protocol, Registry Clearing, Frequency Application	3
Misc Critical	Coherence, Phase Lock, Registry, Accumulation, Vent Path	5

Total: ~100 terms

Level 2: Standard Paper Set (75 terms)

Priority: Sufficient for most standalone papers

Term	Short Definition	Why Essential
D=3	Three spatial dimensions	Axiom 1
S=2	Bilateral symmetry	Axiom 2
** $\mathbb{Q}$ **	Rational substrate	Axiom 3
K-Space	Discrete substrate reality	Where computation occurs
X-Space	Rendered perceptual space	Where humans experience
LERP	K→X interpolation (15.19ms)	Creates qualia
Soliton	Stable substrate pattern	Fundamental entity
Tier	Hierarchical nesting level	0-6 (universe to cell)
J-distance	Parent-child registry depth	7.71 (organism), 8.45 (organ), 9.07 (cell)
N=0 Pivot	Universal root node	All registries terminate here
R (Remainder)	Accumulated processing residue	Health metric
Δ = 19	Healthy baseline remainder	Equilibrium value
R = 69	Malignancy threshold	Critical closure point
Venting	R-clearing to parent/N=0	Health mechanism
α (Alpha)	Vent function (W=1)	Output/clearing
β (Beta)	Torque function (W=2)	Rotation/motion
γ (Gamma)	Socket function (W=3)	Containment/mass
θ (Angle)	Registry pointer deviation	0°=health, 90°=disease
$F(\theta) = \cos(\theta)$	Flow efficiency function	$F(0°)=1$, $F(90°)=0$
90° Phase Lock	Perpendicular blockage	Venting failure
6-9 Hook	Self-sustaining closure at R=69	Cancer mechanism
Toroidal Closure	Donut R-accumulation	Heart failure type

Term	Short Definition	Why Essential
Hexagonal Lattice	2D substrate structure	Geometric basis
Qualia	Subjective experience	LERP output
15.19ms Lag	K→X rendering delay	$J \times S = 7.71 \times 2$
Bilateral	Two-sided (A/B) structure	S=2 manifestation
Q-Operator	A - B difference	Creates qualia
Laminar Coherence	R=19, θ=0° state	Optimal function
Turbulence	R>19, θ>0° state	Dysfunction
Registry	Hierarchical tracking structure	Parent-child chain
Logismos	Discrete K-Space mathematics	Alternative to calculus
VFR Tuple	(Volume, Frequency, Remainder)	K-Space state
Base 32^-1	Substrate counting base	1/32 harmonic
dN	Discrete differential	Not dx
ΣN	Discrete accumulation	Not ∫dx
Φ_in = Φ_out	Equilibrium condition	R maintains at Δ
Accept All/Deny None	Coherence algorithm	Prevents accumulation
α_EM = 1/137.036	Fine structure constant	Derived from geometry
32 Hz	Base harmonic frequency	Substrate clock
40 Hz	Neural/gamma frequency	Alzheimer's treatment
60 Hz	Organ resonance	Therapeutic frequency
Depression (60°)	Partial topple	F=0.5 (50% reduction)
Autism (90°)	Hard metallic lock	Bilateral failure
Alzheimer's (90°)	Soft viscous lock	Temporal loop
Cancer	6-9 hook closure	R=69 pathology
Side A / Side B	Manifold faces	Bilateral structure
Side-B Inversion	180° flip	Autoimmune mechanism
Id-Layer	Fascia/connective tissue	Structural interface
Ib-Layer	Local cellular processes	Molecular level
Registry Pointer	Direction to parent	Angle θ critical
String-8 Closure	Figure-8 fascial tension	Cardiac arrest type
Tier-4	Organism level	J=7.71
Tier-5	Organ level	J=8.45
Tier-6	Cell level	J=9.07
1024-bit	Sovereignty threshold	Maximum coherence
R-Proxy	Biomarker for remainder	HRV, inflammation, etc.
Measurement Arbitration	Reality decides validity	Not theory
Falsification	Tests that prove wrong	Scientific requirement
Derivation	From axioms to conclusion	Not assumption
Sonic Intervention	Frequency therapy	40-60 Hz
Postural Alignment	Vertical θ=0° correction	Opens venting
R-Reduction Protocol	Systematic lowering of R	Sleep, fast, exercise

Term	Short Definition	Why Essential
Topological Closure	Geometric venting blockage	Disease mechanism
Laminar Searching	High-velocity exploration	R=19 enabled
Non-Wanting	No outcome attachment	Enables coherence
Torque (Axial)	Return to axiom focus	Corrects drift
Compilation	Logical consistency check	Framework validity
Theoretical	Not yet validated	Requires testing
Discrete	Countable, not continuous	Substrate property
Rational	$\mathbb{Q}$ (no irrational numbers)	Substrate constraint
Rendering	K→X transform process	Creates experience
Registry Chain	Child→Parent→N=0 path	Information flow
110 Hz	Information carrier	K→X bridge
66 Hz	Matter anchor frequency	Stabilization

Total: 75 terms

Level 3: Compact Introduction (50 terms)

Priority: Minimal set for paper introduction section

Term	One-Line Definition
D=3, S=2, $\mathbb{Q}$	Three axioms: 3D space, bilateral symmetry, rational substrate
K-Space	Discrete geometric substrate where reality computes
X-Space	Continuous rendered experience (what we perceive)
LERP (15.19ms)	Interpolation transforming K→X, creating qualia
Soliton	Stable geometric pattern (particle, cell, organism, etc.)
Tier	Hierarchical level (0=universe, 4=organism, 5=organ, 6=cell)
J-distance	Registry depth (7.71 organism, 8.45 organ, 9.07 cell)
N=0 Pivot	Universal root node of all registries
R (Remainder)	Accumulated processing residue
Δ = 19	Healthy baseline R value
R = 69	Critical threshold for malignant closure
Venting	Clearing R through α-port to parent/N=0
α, β, γ	Three W-functions: vent, torque, socket

Term	One-Line Definition
θ (Angle)	Registry pointer deviation from vertical (0°=health)
F(θ) = cos(θ)	Flow efficiency (F(0°)=1, F(90°)=0, F(60°)=0.5)
90° Phase Lock	Perpendicular blockage stopping venting
6-9 Hook	Self-sustaining closure at R=69 (cancer)
Hexagonal Lattice	2D substrate geometric structure
Qualia	Subjective experience from bilateral integration
Bilateral (S=2)	Two-sided manifold (Side A/B)
Q = A - B	Difference operator creating qualia
Laminar Coherence	State where R=19, θ=0°, Φ_in=Φ_out
Registry	Parent-child tracking chain to N=0
Logismos	Discrete mathematics for K-Space
VFR (Volume, Frequency, Remainder)	K-Space state descriptor
Base 32^-1	Counting system (1/32 = 0.03125)
Φ_in = Φ_out	Equilibrium: input equals output
Accept All/Deny None	Algorithm maintaining R=19
α_EM = 1/137.036	Fine structure constant (derived)
40 Hz, 60 Hz	Therapeutic frequencies (neural, organ)
Depression (60°)	Partial topple, F=50%
Autism (90° hard)	Metallic lock, bilateral failure
Alzheimer's (90° soft)	Viscous lock, temporal loop
Cancer (R=69)	6-9 hook at cellular level
Id-Layer	Fascial/connective substrate interface
Tier-4, Tier-5, Tier-6	Organism, organ, cell levels
1024-bit	Sovereignty threshold (R=19 required)
R-Proxy	Measurable biomarker for R (HRV, inflammation)
Measurement Arbitration	Reality decides via measurements
Falsification	Specific tests that would prove wrong
Sonic Intervention	Frequency therapy (40-60 Hz)
Postural Alignment	Vertical correction to θ=0°
R-Reduction	Lowering R via sleep, fast, exercise, stress reduction
Closure	Topological blockage of venting path
Laminar Searching	Rapid exploration maintaining R=19
Non-Wanting	No attachment to outcomes
Derivation	Logical consequence from axioms
Discrete/Rational	Substrate properties (countable, $\mathbb{Q}$)
Rendering	K→X transformation process
Registry Chain	Path from child through parents to N=0

Total: 50 terms

Level 4: Methods Section Minimum (30 terms)

Priority: Enough to describe CKS methodology in paper

Term	Essential Definition
D=3, S=2, $\mathbb{Q}$	Framework axioms: 3D, bilateral, rational
K-Space / X-Space	Substrate computation / Rendered perception
Soliton	Stable geometric pattern at any tier
Tier (0-6)	Hierarchical nesting (universe→cell)
J-distance	Registry depth (7.71, 8.45, 9.07 for T4-6)
R (Remainder)	Accumulated processing residue
Δ = 19 / R = 69	Healthy baseline / Malignancy threshold
θ (Angle)	Deviation from vertical (0°=health, 90°=disease)
F(θ) = cos(θ)	Flow efficiency function
Venting (α)	R-clearing mechanism
90° Phase Lock	Perpendicular venting blockage
6-9 Hook	R=69 self-sustaining closure
Qualia	Subjective experience (LERP output, 15.19ms)
Bilateral (S=2)	Two-sided manifold, A/B integration
Laminar Coherence	R=19, θ=0° optimal state
Registry Chain	Parent-child path to N=0
Logismos	Discrete K-Space mathematics
VFR Tuple	(Volume, Frequency, Remainder) state
Φ_in = Φ_out	Equilibrium condition
α_EM = 1/137.036	Derived fine structure constant
40 Hz / 60 Hz	Neural / Organ therapeutic frequencies
Depression (60°)	Partial topple, F=0.5
Cancer (R=69)	6-9 hook cellular closure
Id-Layer	Fascial/structural substrate
R-Proxy	Biomarker (HRV, inflammation, metabolic)
Measurement Arbitration	Reality validates via measurements
Falsification	Testable failure conditions
Closure	Topological venting blockage
Non-Wanting	No outcome attachment
Derivation	From axioms via geometry

Total: 30 terms

Level 5: Abstract/Summary (20 terms)

Priority: Minimal context for understanding framework claims

Term	Critical Definition
D=3, S=2, ℚ	Three axioms (3D, bilateral, rational)
K-Space	Discrete substrate (reality)
X-Space	Rendered perception (experience)
Soliton	Stable pattern (particle/cell/organism)
R (Remainder)	Accumulated residue
Δ=19 / R=69	Health baseline / Disease threshold
θ (Angle)	Vertical deviation (0°=health, 90°=disease)
F(θ)=cos(θ)	Flow efficiency
Venting	R-clearing mechanism
90° Lock	Perpendicular blockage
6-9 Hook	Self-sustaining closure
Qualia	Subjective experience (bilateral LERP)
Laminar Coherence	Optimal state (R=19, θ=0°)
Logismos	Discrete mathematics
α_EM=1/137.036	Derived constant
40Hz/60Hz	Therapeutic frequencies
Cancer	R=69 cellular closure
Measurement	Reality validates
Falsification	Testable predictions
Derivation	From axioms

Total: 20 terms

Level 6: Ultra-Compact (15 terms)

Priority: Absolute minimum for any mention of CKS

Term	Minimal Definition
D=3, S=2, ℚ	Framework axioms
K-Space / X-Space	Substrate / Perception
Soliton	Stable pattern
R (Remainder)	Processing residue
R=19 / R=69	Health / Disease thresholds
θ (Angle)	Vertical deviation

Term	Minimal Definition
F(θ)=cos(θ)	Flow efficiency
90° Lock	Venting blockage
6-9 Hook	Malignant closure
Qualia	Subjective experience
Laminar Coherence	Optimal R=19, θ=0°
Logismos	Discrete math
α_EM=1/137.036	Derived constant
Measurement	Validates framework
Derivation	From geometry

Total: 15 terms

Level 7: Core Concepts (10 terms)

Priority: Skeleton framework understanding

Term	Core Definition
D=3, S=2, ℚ	Three axioms
K-Space	Discrete substrate
Soliton	Stable pattern
R (Remainder)	Health metric
θ (Angle)	0°=health, 90°=disease
F(θ)=cos(θ)	Efficiency function
R=69	Disease threshold
Qualia	Experience mechanism
Logismos	Substrate mathematics
Derivation	Axiom→measurement

Total: 10 terms

Level 8: Absolute Minimum (7 terms)

Priority: Cannot reduce further without losing coherence

Term	Irreducible Definition
D=3, S=2, ℚ	Framework axioms (3D, bilateral, rational substrate)
K-Space	Discrete geometric substrate where reality computes

Term	Irreducible Definition
R (Remainder)	Accumulated residue (Δ=19 health, R=69 disease)
θ (Angle)	Registry deviation (0° optimal, 90° pathological)
Soliton	Stable geometric pattern (all entities)
Derivation	Geometric consequence from axioms
Measurement	Reality validation (not theory)

Total: 7 terms

Level 9: Emergency Ultra-Minimal (5 terms)

Priority: Absolute bare minimum (loses significant explanatory power)

Term	Ultra-Minimal
D=3, S=2, $\mathbb{Q}$	Three geometric axioms
K-Space	Discrete substrate
R, θ	Health metrics (remainder, angle)
Derivation	From axioms
Measurement	Validates predictions

Total: 5 terms

Warning: At this level, framework barely comprehensible. Use only when space is severely constrained (e.g., tweet, title, caption).

Level 10: Theoretical Absolute Minimum (3 terms)

Priority: Only if forced to explain in single sentence

Term	Absolute Minimum
D=3, S=2, $\mathbb{Q}$	Axioms
Derivation	Geometry forces consequences
Measurement	Reality validates

Total: 3 terms

Critical: At this level, framework is essentially unexplained. Use only as

teaser/reference to full paper.

USAGE GUIDE

Selecting Appropriate Level

Context	Recommended Level	Term Count
Full research paper	Level 1-2	75-100
Journal article	Level 3	50
Conference abstract	Level 5	20
Poster presentation	Level 6	15
Tweet/social media	Level 8-9	5-7
Figure caption	Level 9	5
Title/teaser	Level 10	3
Methods section	Level 4	30
Introduction section	Level 3-4	30-50
Discussion section	Level 3	50
Supplementary materials	Level 0-1	100-200

Expansion Strategy

When space allows, expand systematically:

1. **Start at constraint level** (e.g., abstract = Level 5, 20 terms)
2. **Add next level terms** as space permits
3. **Prioritize by relevance** to specific paper topic
4. **Always include**: Axioms, K/X-Space, R, θ, measurement
5. **Domain-specific additions**: Add disease terms for medical papers, engineering for technical, etc.

Customization by Domain

For Physics Papers

Add from Level 1-2: α_EM, constants, discrete/rational, hexagonal lattice, frequencies

For Biology Papers

Add from Level 1-2: Disease terms (cancer, Alzheimer's, autism), Id/Ib-layers, tiers, R-proxies

For Mathematics Papers

Add from Level 1-2: Logismos, VFR, base 32^-1, dN, ΣN, discrete operators

For Clinical Papers

Add from Level 1-2: R-reduction protocol, sonic intervention, postural alignment, R-proxies, topological diagnosis

For Consciousness Papers

Add from Level 1-2: Qualia, LERP, 15.19ms, bilateral, Q-operator, Side A/B, rendering

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Critical Rules

Never omit:

1. **Axioms (D=3, S=2, $\mathbb{Q}$)** - Foundation of everything
2. **K-Space** - Where framework operates
3. **Measurement** - Validation method
4. **Derivation** - How conclusions reached

Always include when discussing:

- Health: R, Δ=19, θ, venting
- Disease: R=69, closures, 90° locks
- Treatment: Frequencies, posture, R-reduction
- Consciousness: Qualia, LERP, bilateral
- Mathematics: Logismos, discrete, VFR

Term Introduction Order

Optimal sequence for first mention:

1. **Axioms** (D=3, S=2, $\mathbb{Q}$)
2. **K-Space** (substrate)

3. **Soliton** (patterns)
4. **R** (health metric)
5. **θ** (angle metric)
6. **Derivation** (method)
7. **Measurement** (validation)
8. Then domain-specific terms as needed

Reduction Methodology

How these levels were created:

1. **Start with complete set** (LEX-1, 200+ terms)
2. **Score each term** by:
    - Frequency across 257 papers
    - Explanatory necessity (how many concepts depend on it)
    - Measurement linkage (connects to validation)
    - Axiom proximity (how directly derived)
3. **Remove lowest-scoring** at each level
4. **Verify comprehensibility** - can framework still be understood?
5. **Iterate** until minimum reached
6. **Final check**: Level 8 (7 terms) is absolute minimum for coherence

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Quality Check Questions

Before submitting paper, ask:

1. Have I introduced D=3, S=2, $\mathbb{Q}$?
2. Have I explained K-Space vs X-Space?
3. If discussing health, have I defined R and θ?
4. If claiming derivation, have I shown measurement match?
5. If proposing treatment, have I stated it's theoretical?
6. Is every CKS term I use defined somewhere in paper?
7. Could reader understand my claims with only terms I've introduced?

END CKS-LEX-2-2026

Status: Complete Hierarchical Lexicon Series

Levels: 11 (from 200+ terms down to 3)

Purpose: Scalable terminology for any space constraint

Validation: Each level independently comprehensible

Usage:

- Select level based on space available
- Customize by domain as needed
- Always include axioms, K-Space, measurement, derivation
- Expand systematically when space permits

The 7-term minimum (Level 8) is irreducible.

Below that, framework becomes incomprehensible.

Vented. Scaled. Complete.

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026

[**CKS-MATH-13-2026**] The Resonant Epoch. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-13-2026

[**CKS-MATH-16-2026**] The Geometric Necessity of 32. Github: https://github.com/ghowland/cks/tree/main/papers/MAT
MATH-16-2026

[**CKS-DWDM-5-2026**] The Geometry of the 66th Harmonic. Github: https://github.com/ghowland/cks/tree/main/papers/I
DWDM-5-2026

[**CKS-MATH-17-2026**] The 7-Bubble Jacobian. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-17-2026

[**CKS-MATH-18-2026**] The 84-Bit Word on a 32-Bit Computer. Github: https://github.com/ghowland/cks/tree/main/pape
MATH-18-2026

[**CKS-MATH-19-2026**] Topological Recirculation. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CK
MATH-19-2026

[**CKS-MATH-20-2026**] The Winding Torus. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-20-2026

[**CKS-MATH-21-2026**] The Final Constant Closure. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/C
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